PROJECT CASCADE

cas·cade (n.) — water falling from a steep height; the exact overflow this system is built to prevent.

SMART TANK MONITORING AND AUTOMATIC PUMP CONTROL SYSTEM

IoT MONITORING | AUTOMATIC CONTROL | LIVE DASHBOARD | WATER CONSERVATION

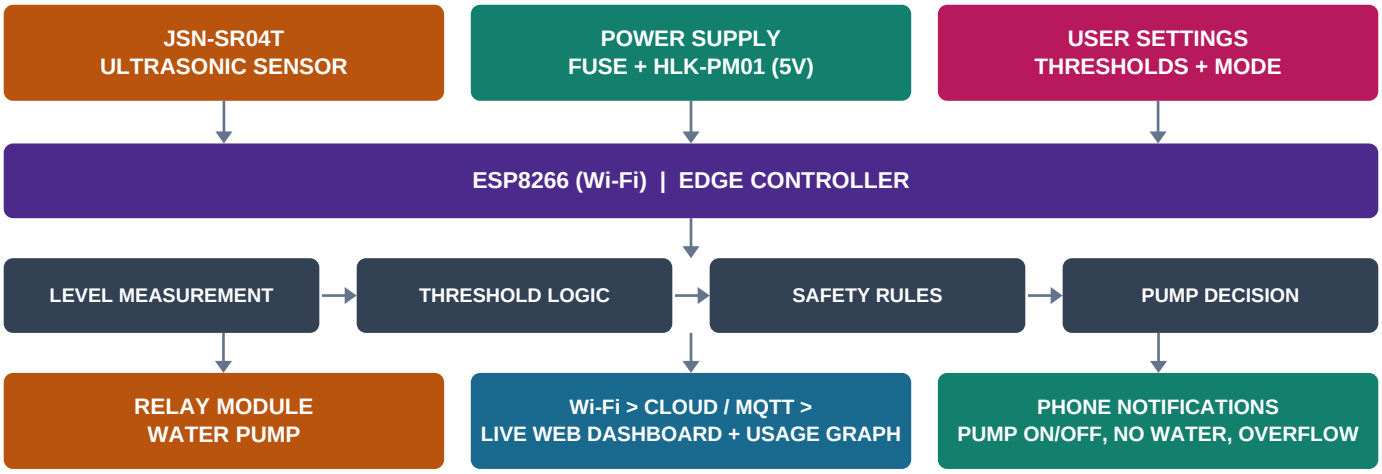
1 PROJECT CONCEPT

Overhead water tanks in homes, hostels and apartments are usually filled by pumps switched by hand. With no view of the level, tanks overflow and waste water and electricity, or run dry and damage the pump. Project Cascade is a low-cost IoT tank controller that uses an ESP8266 to measure the water level, show it on a live web dashboard, and switch the pump automatically or manually.

2 METHODOLOGY

A waterproof JSN-SR04T ultrasonic sensor on the tank lid measures the distance to the water surface. After one-time calibration, the ESP8266 takes the median of five readings and converts it to level percentage and litres. Level and pump status go over Wi-Fi to a live web dashboard that shows the current level and a graph of water usage over time, where the user sets start and stop thresholds. **In automatic mode**, the ESP8266 turns the pump ON through an opto-isolated relay at the start level and OFF at the stop level. **In manual mode**, the pump is controlled from the website and phone notifications report pump on/off, no water and overflow. The control logic runs on the ESP8266 itself, so it keeps working if Wi-Fi fails. Safety rules stop the pump on a maximum run time, sensor faults or dry running, and the relay stays OFF at boot. Power comes from the mains through a fuse/MCB and an HLK-PM01 module, all inside an IP65 enclosure.

3 PROPOSED HARDWARE BLOCK DIAGRAM



4 PROPOSED OUTCOME

Project Cascade aims to show that a simple, affordable device can keep a tank filled just right without anyone watching it. It is expected to prevent overflow and dry running, cut water and electricity wastage, protect the pump, and give users a live level reading, a usage graph and alerts. Built from components costing under Rs. 2,000, it can be fitted to any existing tank and pump.